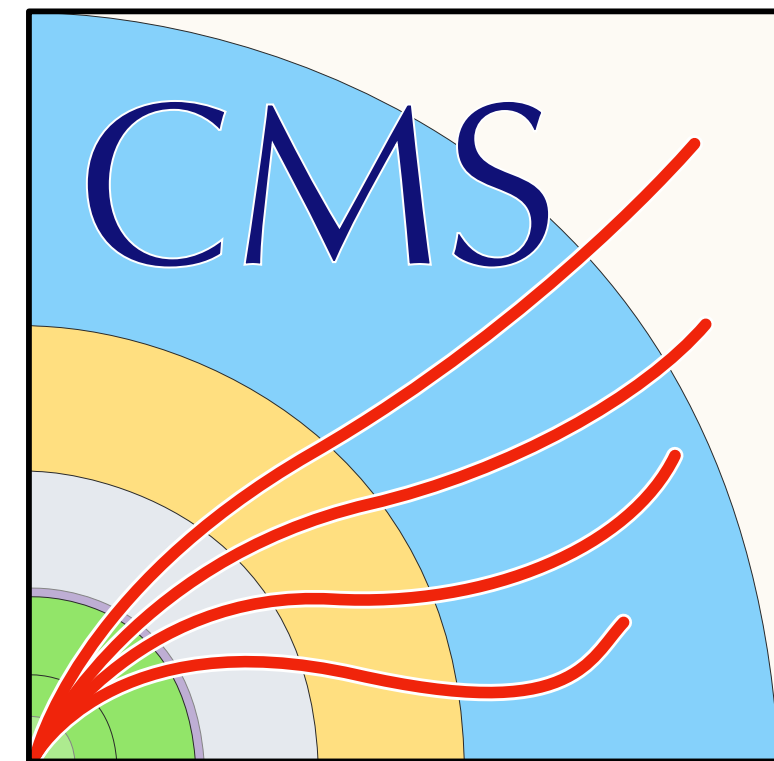




Off-shell Higgs boson Production Measurements

Mohit Srivastav

Johns Hopkins University, CMS Joint 4-Lepton Study Team (CJLST)



Objectives

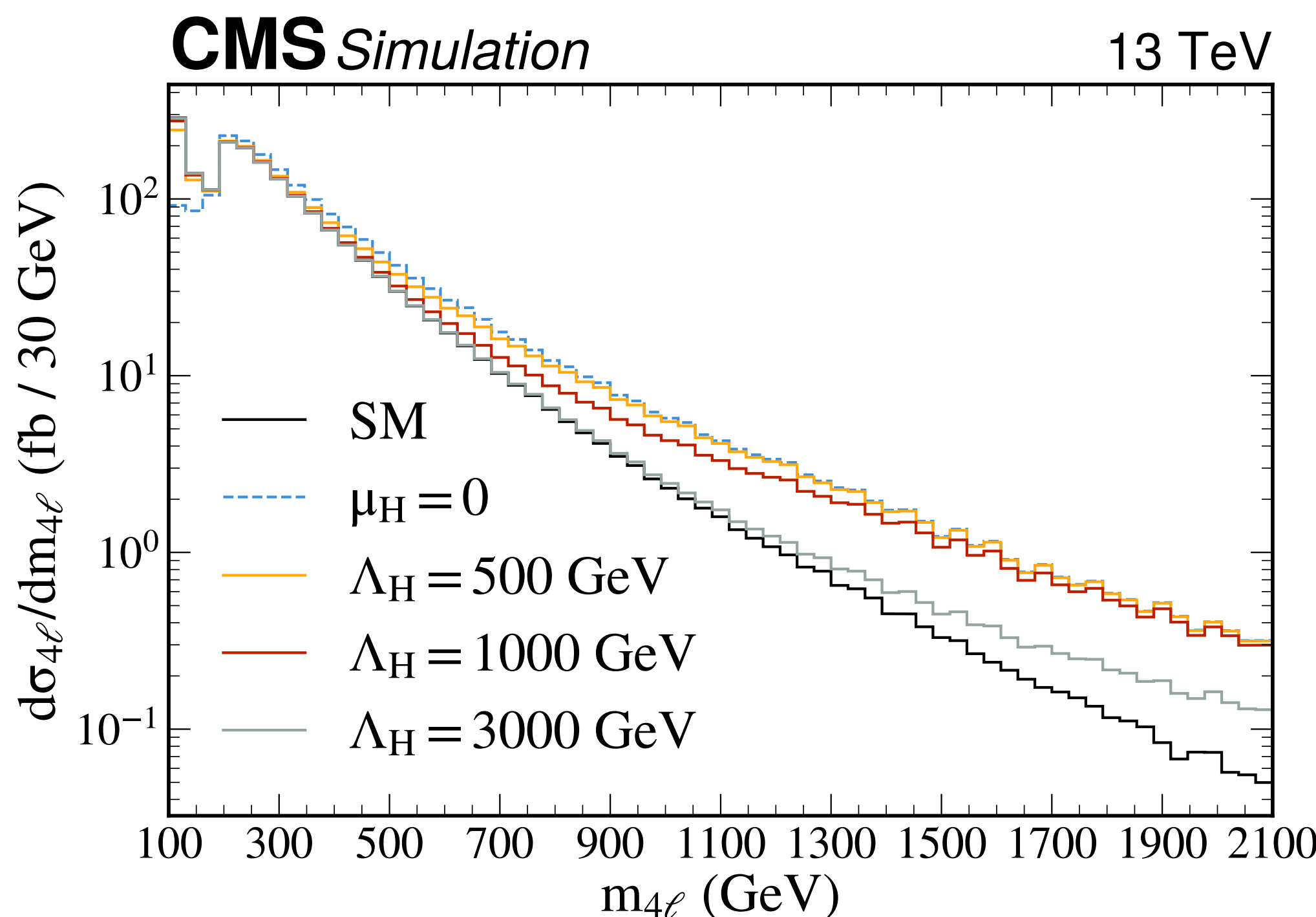
Multiple measurements of off-shell Higgs boson production are presented:

- 1 The first test of Higgs substructure on CMS
- 2 The first measurement of κ_λ in the off-shell region
- 3 The most precise indirect measurement of light-Yukawa couplings to date
- 4 Measurements of BSM phenomena in the EFT limit in gluon-fusion production
- 5 Constraints on the width with both heavy and light terms floated
- 6 A combination of all the Run-2 off-shell width measurements to date

The Off-shell Higgs boson

The off-shell region of the Higgs boson is present in VV decays above $2m_V$, and is characterized by a strong interference component with background. The hallmark of the off-shell region is the independence from the the width, as $4 \text{ MeV} \ll 2m_V$.

$$\frac{\sigma(i \rightarrow H \rightarrow f)}{ds} \propto \frac{1}{(s - M_H^2)^2 + M_H^2 \Gamma_H^2} \quad (1)$$

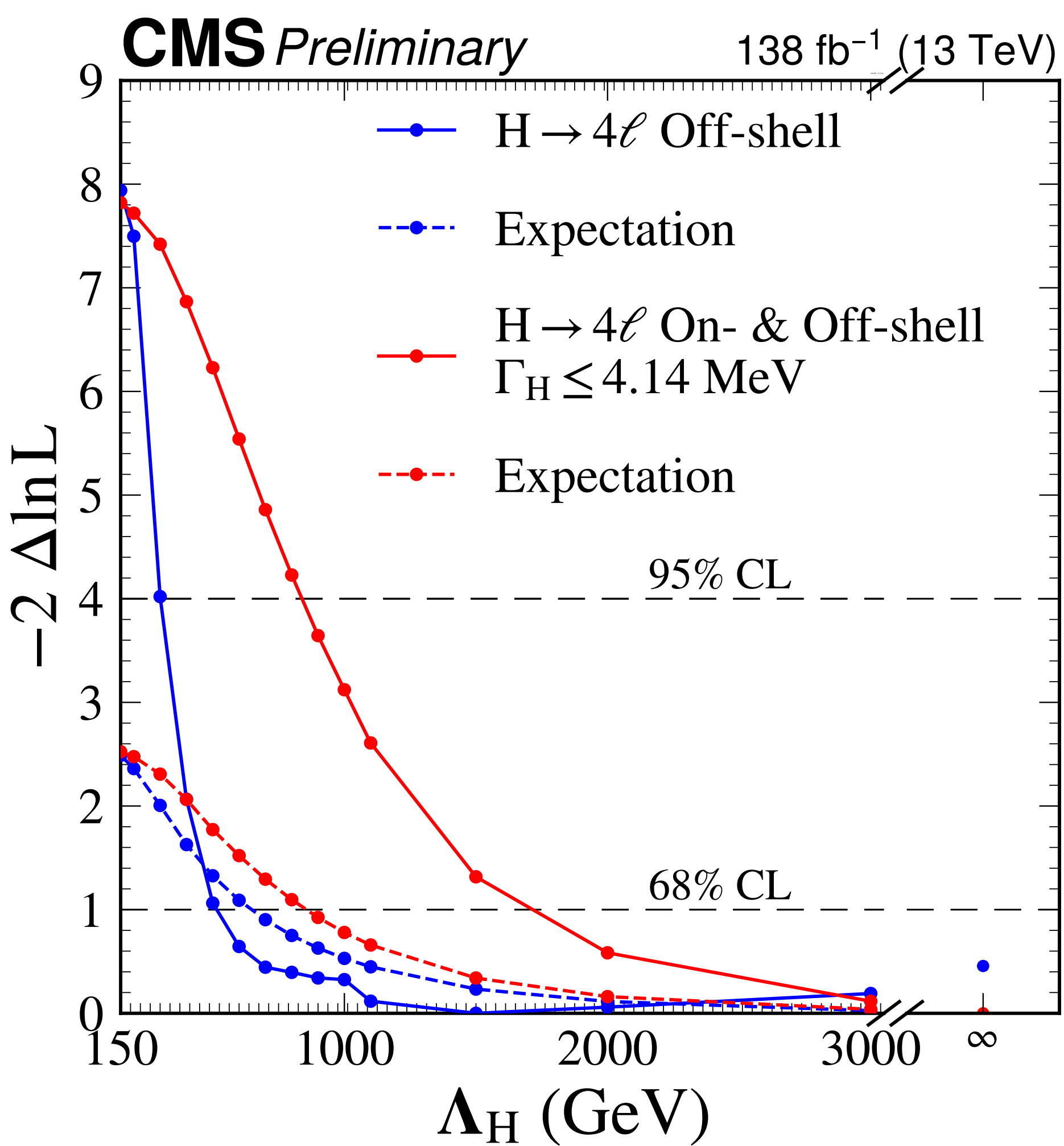


Observables

Observables crafted using MELA and MiLoMerge in tandem with $m_{4\ell}$

Higgs Substructure

$$F(q^2) = \left[\frac{1}{1 + |q^2|/\Lambda_H^2} \right]^2, \quad d = \frac{\hbar c}{\Lambda_H}$$



Results

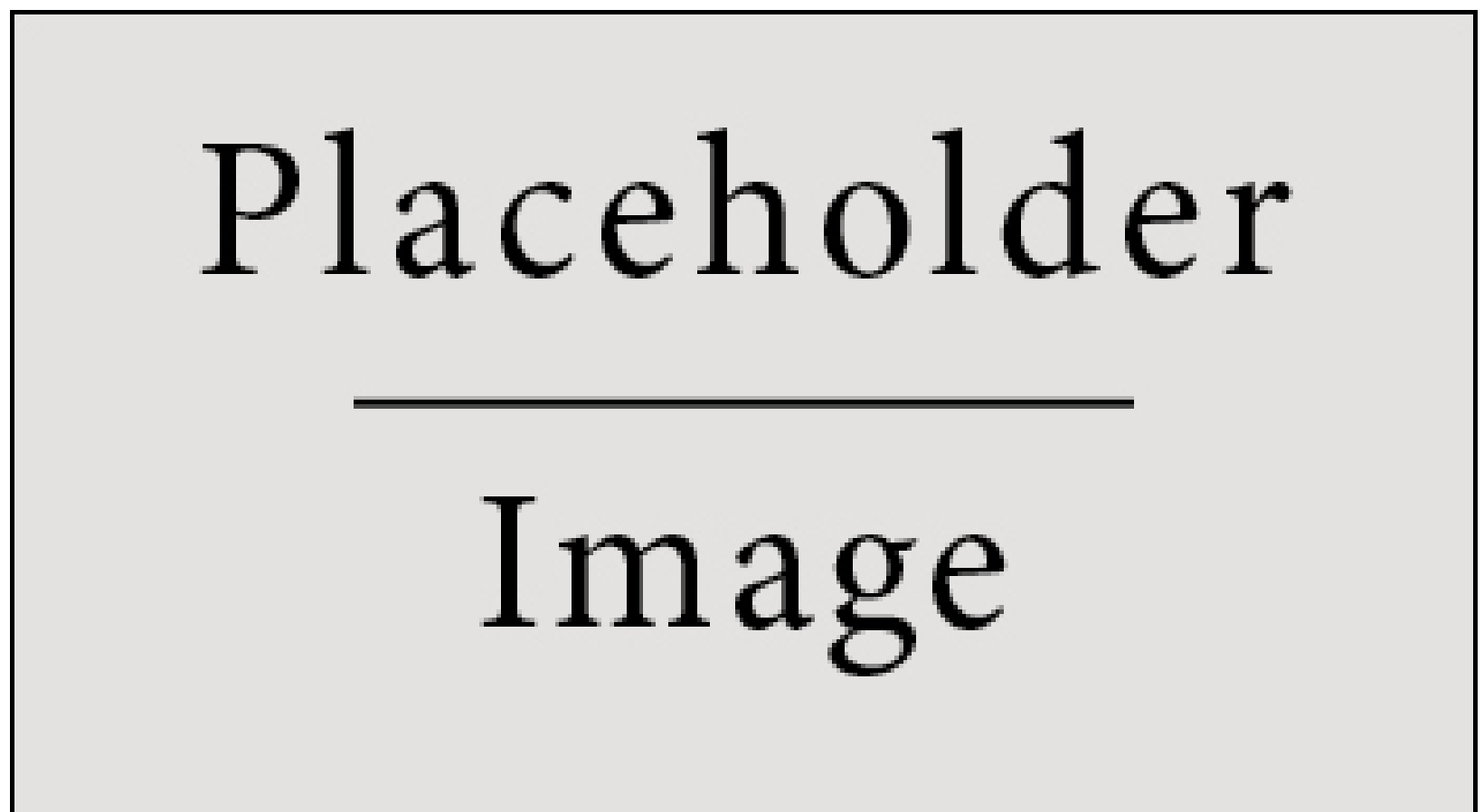
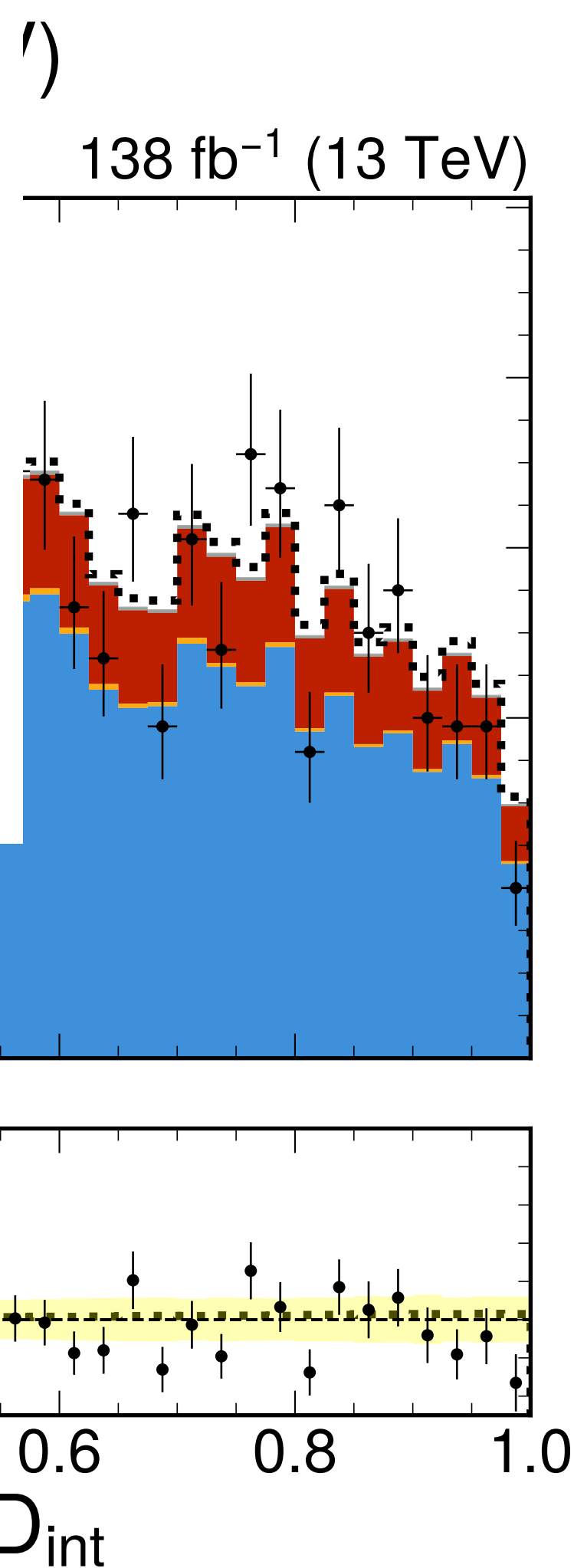
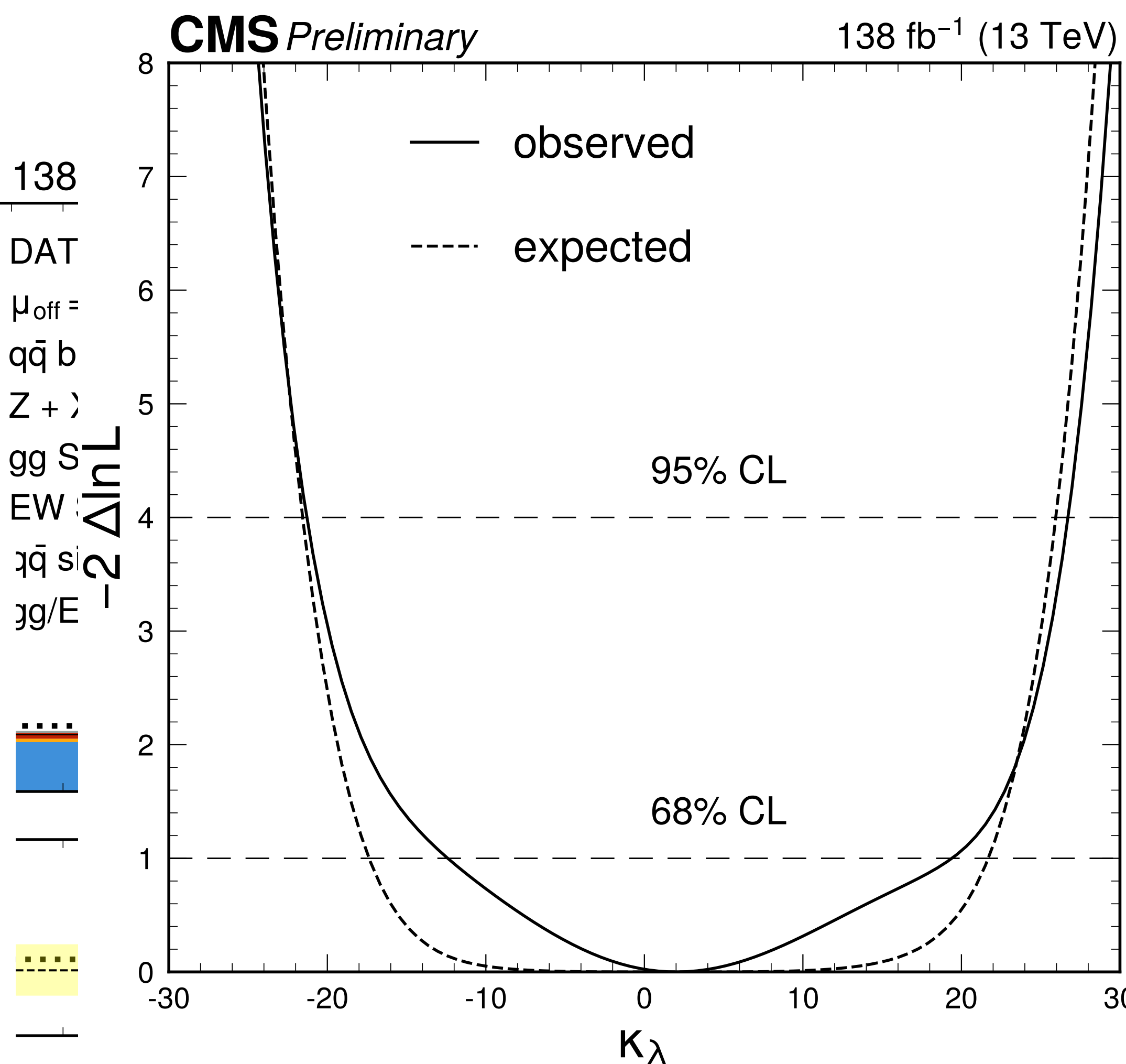


Figure 1:Figure caption

Yukawa Couplings



Conclusion

Nunc tempus venenatis facilis. **Curabitur suscipit** consequat eros non porttitor. Sed a massa dolor, id ornare enim. Fusce quis massa dictum tortor **tincidunt mattis**. Donec quam est, lobortis quis pretium at, laoreet scelerisque lacus. Nam quis odio enim, in molestie libero. Vivamus cursus mi at *nulla elementum sollicitudin*.

Additional Information

Maecenas ultricies feugiat velit non mattis. Fusce tempus arcu id ligula varius dictum.

- Curabitur pellentesque dignissim
- Eu facilis est tempus quis
- Duis porta consequat lorem

References

- [1] J. M. Smith and A. B. Jones.
Book Title.
Publisher, 7th edition, 2012.
- [2] A. B. Jones and J. M. Smith.
Article Title.
Journal title, 13(52):123–456, March 2013.

Acknowledgements

Nam mollis tristique neque eu luctus. Suspendisse rutrum congue nisi sed convallis. Aenean id neque dolor. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Contact Information

- Web: <http://www.university.edu/smithlab>
- Email: john@smith.com
- Phone: +1 (000) 111 1111

PLACEHOLDER
LOGO

PLACEHOLDER
LOGO

Nunc tempus venenatis facilis. Curabitur suscipit consequat eros non porttitor. Sed a massa dolor, id ornare enim: